AI Note

This document outlines how Artificial Intelligence was utilized during the development and execution of the Capacity Forecaster project.

What AI Helped With

Code Generation & Scaffolding

AI-assisted development tools were used to generate boilerplate code for React frontend components, FastAPI backend routes, SQLite database integration, authentication workflows, forecasting modules, and visualization components.

Authentication Module Development

AI assistance was used during the implementation of user registration, login functionality, JWT token authentication, password hashing using bcrypt, and SQLite database operations.

Natural Language Query Processing

Google Gemini AI Studio API was integrated to interpret user capacity-planning questions such as:

"When will disk usage reach 80%?"

The AI extracts the required metric, threshold value, and forecasting intent from natural language input.

Forecast Analysis

AI-assisted development supported the implementation of forecasting models using Linear Regression and ARIMA to predict future CPU, Memory, and Disk utilization trends.

Recommendation Generation

Gemini AI generates business-oriented recommendations based on forecast results, helping users understand capacity risks and plan infrastructure improvements proactively.

Documentation & Testing Support

AI tools assisted in generating test cases, debugging backend APIs, improving validation logic, optimizing database operations, and preparing project documentation.

What AI Got Wrong (and How We Fixed It)

Incorrect Metric Identification

Occasionally the AI selected an incorrect resource metric from user queries.

Fix: Added validation checks to ensure only supported metrics (CPU, Memory, Disk) are processed.

Invalid Forecast Suggestions

Some AI-generated forecasting approaches produced unrealistic predictions.

Fix: Forecast results were validated using historical utilization data and reviewed before integration.

Authentication Logic Suggestions

AI-generated authentication snippets occasionally required modification to align with JWT security standards.

Fix: Authentication workflows were manually reviewed and tested before deployment.

API Service Interruptions

Gemini API requests occasionally experienced temporary failures or rate limits.

Fix: Exception handling and fallback responses were implemented to maintain system stability.

Best Prompts Used

Natural Language Query Prompt

"Extract the resource metric and threshold from the user's forecasting question."

Forecast Recommendation Prompt

"Based on forecast results, provide a short business recommendation and risk assessment."

Capacity Planning Prompt

"Analyze forecast trends and suggest actions to prevent resource utilization from exceeding critical thresholds."

User Assistance Prompt

"Answer user questions related to resource forecasting, risk levels, and capacity planning."

AI Tools Used

Tool	Purpose
Antigravity AI Code Assistant	Code generation, debugging, optimization, and development assistance
Google Gemini AI Studio API	Natural language query processing and recommendation generation
SQLite	User data and query history storage
JWT Authentication	Secure user session management
Linear Regression / ARIMA	Resource utilization forecasting
Pandas	Data processing and analysis

Human Validation Statement

All AI-generated code, recommendations, authentication logic, forecasts, and outputs were reviewed, tested, validated, and refined by the development team before inclusion in the final project deliverable.

AI Tools Link

Anitygravity - https://antigravity.google/

Google AI Studio - https://aistudio.google.com/

ChatGPT - https://chatgpt.com/